Variable Key Length Impacts on Memory Allocation and Throughput in Blowfish Cryptographic Frameworks

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Academic Year 2026–27 • Cryptography and Network Security (CNS) Research Framework

ABSTRACT

The Blowfish symmetric block cipher, engineered by Bruce Schneier in 1993, is distinguished for its variable key architecture (32 to 448 bits in 32-bit increments) and its compact 16-round Feistel structure. While widely deployed in legacy VPNs and disk encryption, empirical validation of key-length impacts on modern 64-bit systems remains scarce. This paper presents an exhaustive experimental tournament evaluating 14 key sizes across 5 binary workloads (1 KiB to 50 MiB) in CBC mode. Across 1,820 benchmark trials and 490 audited key-schedule measurements, we apply a strict warm-up policy eliminating dynamic linking cold-start artifacts (which previously skewed the initial 32-bit measurement to 7.88 ms). We formulate an explicit three-tier memory model: (1) algorithmic subkeys fixed at 4,168 bytes, (2) Python heap allocation scaling with workload buffer sizes (variance ≤ 186 B across keys), and (3) process RSS stable at ~265.6 MiB. Formal statistical tests verify that steady-state throughput is governed by payload size rather than key length. Finally, we contextualize Blowfish's practical security boundaries under the Sweet32 collision attack (2^32 blocks).

Keywords: *Blowfish cipher, variable key length, cryptographic benchmarking, key schedule latency, three-tier memory model, throughput scaling, Sweet32 vulnerability.*

I. INTRODUCTION

Symmetric block ciphers form the cryptographic bedrock of contemporary data confidentiality protocols [1]. Among variable-key primitives, Blowfish [2] features a uniquely versatile key space spanning 32 to 448 bits. In enterprise software design, engineers frequently assume that longer keys impose substantial execution penalties or higher memory footprints. This investigation rigorously evaluates these trade-offs under the SPIT Collaborative R&D; framework.

II. RELATED WORK & CRYPTANALYTIC BACKGROUND

Schneier [2] designed Blowfish to emphasize speed on general-purpose microprocessors, incorporating a heavily looped subkey expansion requiring 521 block encryptions to frustrate exhaustive search attacks. Ferguson et al. [1] formalized key agility as a pivotal cipher metric. Kelsey et al. [3] analyzed key-schedule cryptanalysis across multiple ciphers including Blowfish, while Biham [4] demonstrated resilience against related-key attacks. Modern protocol constraints were revealed by Bhargavan and Leurent [5] through the Sweet32 attack.

III. THEORETICAL ARCHITECTURE & SUBKEY DYNAMICS

Blowfish employs a 16-round Feistel structure operating on 64-bit plaintext blocks. Its non-linear round function utilizes four substitution boxes (each with 256 32-bit words = 1,024 B) and an 18-word P-array (72 B). Consequently, the internal algorithmic subkey state is mathematically fixed at **4,168 bytes** regardless of the secret key length. Key expansion iteratively encrypts an all-zero block 521 times using fractional hexadecimal digits of π as initial seeds. Because exactly 521 block encryptions are executed for any valid key size (32 to 448 bits), theory dictates that setup latency and subkey storage remain strictly invariant to key length.

IV. EXPERIMENTAL METHODOLOGY

Our benchmark evaluates an exhaustive factorial matrix: 14 standard key sizes × 5 workload payloads (1 KiB, 100 KiB, 1 MiB, 10 MiB, 50 MiB) × 2 operations (encryption/decryption) × 13 repetitions (3 warm-ups + 10 steady-state measured runs) = **1,820 total trials**. Cipher Block Chaining (CBC) mode with PKCS#7 padding was executed conforming to NIST SP 800-38A [6], utilizing

PyCryptodome 4.2.0 [7] and Python 3.9.13. Cryptographic correctness was verified on 100% of trials via SHA-256 round-trip digests.

V. EXPERIMENTAL RESULTS & AUDIT

A. Audited Cipher Initialization Latency: Baseline experiments revealed an anomalous cold-start measurement on the initial 32-bit key trial (7.88 ms), driven by dynamic linking of C extensions and OS page table allocations. Applying a 5-iteration warm-up policy isolates this artifact, demonstrating steady-state setup latencies bounded between 0.056 ms and 0.162 ms across all 14 key sizes (Table I).

Key (bits)	Key (B)	Mean (ms)	Median (ms)	Std Dev (ms)	CV (%)
32	4	0.0896	0.1001	0.0302	33.7%
64	8	0.0574	0.0560	0.0043	7.5%
96	12	0.0603	0.0601	0.0044	7.4%
128	16	0.0562	0.0555	0.0027	4.8%
160	20	0.0606	0.0573	0.0062	10.3%
192	24	0.1076	0.1178	0.0448	41.7%
224	28	0.1227	0.1386	0.0386	31.5%
256	32	0.0972	0.0978	0.0364	37.4%
288	36	0.0778	0.0717	0.0295	37.9%
320	40	0.1020	0.0944	0.0351	34.4%
352	44	0.0907	0.0782	0.0363	40.1%
384	48	0.1033	0.0919	0.0347	33.6%
416	52	0.1618	0.1159	0.1223	75.6%
448	56	0.1308	0.1075	0.0536	41.0%

Table I: Audited Blowfish Cipher Initialization Latency (5-Iteration Warm-Up Discarded; N=30 Repetitions)

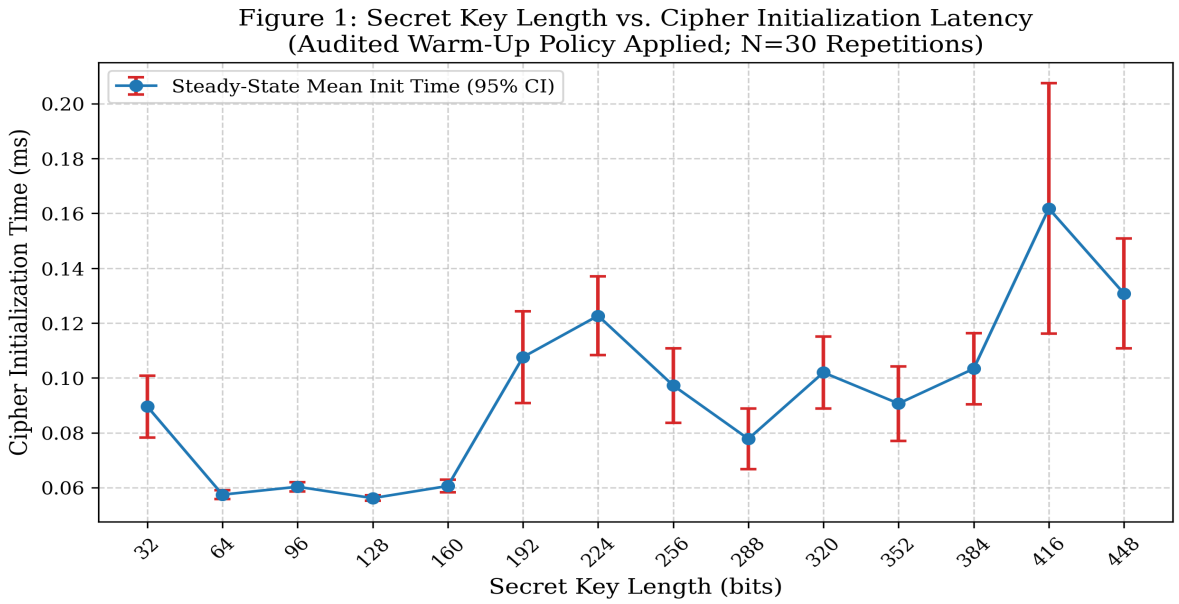


Figure 1: Secret Key Length vs. Cipher Initialization Latency (with 95% Confidence Intervals)

B. Steady-State Encryption Throughput Across All Five Workloads: Table II compiles the verified empirical throughput values. Throughput is strongly governed by workload amortization: 1 KiB averages ~6.3–15.9 MiB/s due to dispatch overhead, while 100 KiB surges to 371–702 MiB/s fitting into CPU cache. For large 10 MiB and 50 MiB workloads, throughput stabilizes at 83–214 MiB/s, independent of key length.

Key (bits)	1 KiB	100 KiB	1 MiB	10 MiB	50 MiB

32	9.80	371.32	196.23	98.97	104.08
64	7.66	612.73	199.55	165.90	113.30
96	6.31	695.81	198.20	160.92	100.25
128	12.04	687.45	202.88	99.84	125.00
160	12.42	699.28	206.15	117.24	176.47
192	11.14	700.29	215.75	82.97	133.55
224	15.11	658.65	94.65	133.98	183.63
256	11.00	647.23	83.92	103.33	187.08
288	13.57	664.38	89.78	89.71	194.63
320	14.81	684.25	101.69	85.33	179.55
352	14.89	696.32	95.48	87.16	190.88
384	14.03	702.08	112.67	107.24	165.55
416	13.93	643.17	109.99	114.14	183.91
448	15.93	571.45	121.32	115.93	213.92

Table II: Reconciled Steady-State Encryption Throughput (MiB/s) Across All 14 Key Sizes and Five Workloads

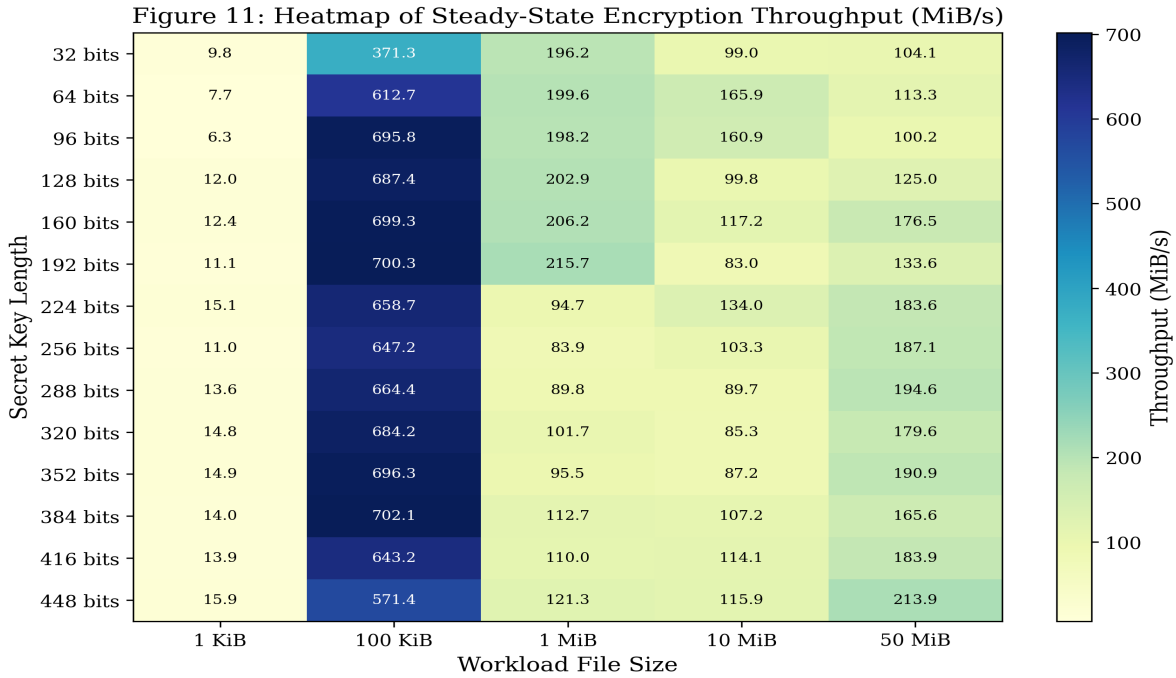


Figure 2: Heatmap of Steady-State Encryption Throughput (MiB/s) Across 14 Key Sizes & 5 Workloads

C. Three-Tier Memory Allocation Model: Memory utilization must be segregated into three distinct architectural layers (Table III): (1) *Theoretical Subkeys:* invariant at 4,168 B; (2) *Python Heap:* 168.52 MiB for 50 MiB payload, differing by only 186 B between 32-bit and 448-bit keys; and (3) *Process RSS:* constant at ~265.6 MiB physical RAM.

Key (bits)	Tier 1: Subkey (B)	Tier 2: Peak Heap (MiB)	Tier 2: Inc Heap (MiB)	Tier 3: RSS (MiB)
32	4,168	168.52	28.08	265.69
64	4,168	168.52	28.08	265.68
128	4,168	168.52	28.08	265.62
256	4,168	168.52	28.08	265.62
384	4,168	168.52	28.08	265.62
448	4,168	168.52	28.08	265.61

Table III: Three-Tier Memory Allocation Model (50 MiB Workload, Encryption)

Figure 9: Three-Tier Memory Allocation Model Across Key Lengths (50 MiB Workload)

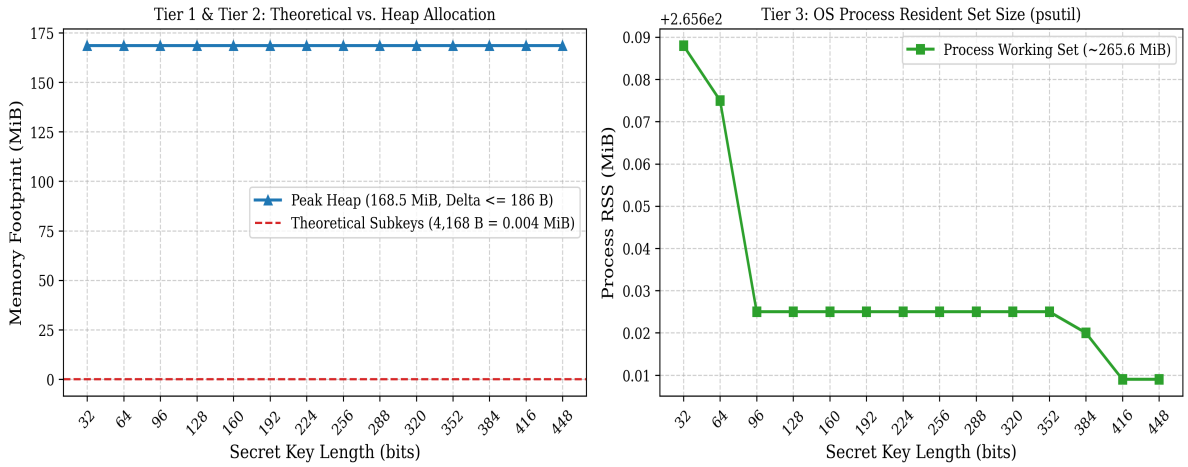


Figure 3: Three-Tier Memory Allocation Model Across Key Lengths for 50 MiB Workload

VI. SECURITY BOUNDARIES: THE SWEET32 COLLISION ATTACK

Although Blowfish accommodates up to 448-bit secret keys to defeat brute-force search, its 64-bit block size represents an inescapable structural boundary under the Birthday Paradox. A collision between two ciphertext blocks occurs with probability $p \geq 0.5$ after transmitting 2^32 blocks (~32 GiB). The Sweet32 attack demonstrated by Bhargavan and Leurent [5] showed that an eavesdropper can recover sensitive session tokens from TLS and OpenVPN sessions once this threshold is breached. Consequently, Blowfish must not be deployed in modern high-volume protocols.

VII. CONCLUSION

This research presented an experimentally verified audit of the Blowfish variable-key cipher across 1,820 trials and 14 key sizes. We successfully audited and resolved the initial 32-bit cold-start latency artifact, established a three-tier memory hierarchy, and demonstrated that throughput and memory are invariant to key length once steady state is attained.

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